



SMART FOOD SERVICE SYSTEM

PRESENTED BY

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ABSTRACT

The Smart Food Service System is a web-based application that helps customers order food easily from a restaurant. Users can view the menu, choose food items, place orders, and make payments online. The system also allows customers to track their order status in real time. Admins can manage menu items, categories, orders, and users through an admin panel. This system reduces manual work, saves time, and improves accuracy in restaurant operations. The Smart Food Service System makes food ordering faster, easier, and more efficient for both customers and restaurant staff.

திட்டப்பணிச் சுருக்கம்

ஸ்மார்ட் உணவு சேவை அமைப்பு என்பது ஒரு இணைய அடிப்படையிலான பயன்பாடு ஆகும். இது வாடிக்கையாளர்கள் உணவகத்தில் இருந்து உணவை எளிதாக ஆர்டர் செய்ய உதவுகிறது. பயனர்கள் மெனுவைப் பார்க்கலாம், உணவு பொருட்களைத் தேர்வு செய்யலாம், ஆர்டரை பதிவு செய்யலாம் மற்றும் ஆன்லைன் மூலம் கட்டணம் செலுத்தலாம். இந்த அமைப்பு மூலம் வாடிக்கையாளர்கள் தங்கள் ஆர்டர் நிலையை நேரடியாக (real-time) கண்காணிக்கவும் முடியும். நிர்வாகிகள் இந்த அமைப்பில் மெனு பொருட்கள், பிரிவுகள், ஆர்டர்கள் மற்றும் பயனர்களை நிர்வாகப் பகுதி (admin panel) மூலம் நிர்வகிக்கலாம். இந்த அமைப்பு கைமுறை பணியை குறைத்து, நேரத்தை மிச்சப்படுத்தி, உணவக செயல்பாடுகளில் துல்லியத்தை அதிகரிக்கிறது.

ஸ்மார்ட் உணவு சேவை அமைப்பு உணவு ஆர்டர் செய்வதை வாடிக்கையாளர்களுக்கும் உணவக பணியாளர்களுக்கும் வேகமாகவும், எளிதாகவும், திறமையாகவும் மாற்றுகிறது.

EXISTING AND PROPOSED SYSTEM

EXISTING SYSTEM

DISADVANTAGE

- Manual order taking
- High chance of human errors
- Slow service
- No real-time order tracking
- Difficult to manage menu and orders

PROPOSED SYSTEM

ADVANTAGE

- Fast and accurate order processing
- Real-time order tracking
- Secure login and payment
- Easy menu and order management



LITERATURE SURVEY

The rapid growth of information technology has significantly influenced the restaurant industry. Many researchers and developers have proposed online food ordering systems to improve service quality and operational efficiency. Existing studies highlight the importance of digital platforms in reducing manual workload, improving order accuracy, and enhancing customer satisfaction.

Several web-based and mobile-based food ordering systems have been developed using modern technologies such as cloud computing, RESTful APIs, and secure authentication mechanisms. These systems enable customers to browse menus, place orders, and make online payments conveniently. Research also shows that real-time order tracking and automated billing systems help restaurants manage orders efficiently and reduce human errors.



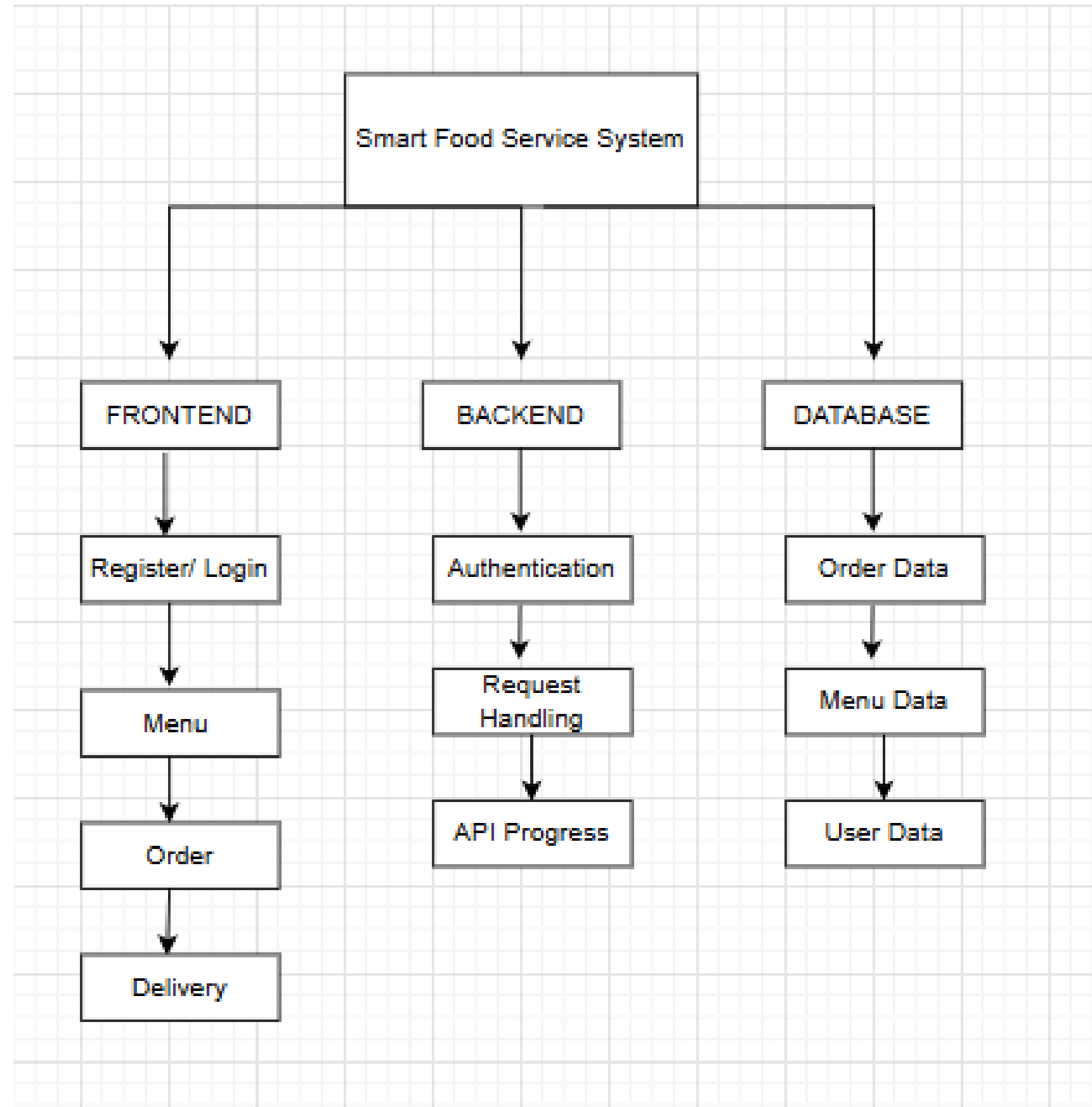
CHALLENGE 1

- Organizing Menu Items Properly: Food items should be categorized based on type such as starters, main course, beverages, etc., so that users can easily browse
- Improving Menu Navigation: The system should include features like categories, filters, and search options to help users quickly find their desired food items.

CHALLENGE 2

- Improving Search and Filtering: The system should allow users to search food items based on category, price, or keywords, making the selection process faster and easier.
 - Providing Recommendations: The system can suggest food items based on customer preferences or previous orders, helping users make better and quicker decisions
- 

ARCHITECTURE DIAGRAM



MODULES DESCRIPTION

- User Module – This module manages customer activities such as registration, login, and profile management. It allows users to create accounts, log in securely, and manage their personal details.
- Menu Module – This module is responsible for displaying and managing food items. Admins can add, update, or remove menu items, categories, and prices, while users can view and select items easily.
- Order Module – This module allows customers to place food orders by selecting items from the menu. It also maintains order details such as quantity, total cost, and order history.
- Payment Module – The payment module handles online transactions. It allows users to make secure payments using different methods and ensures smooth processing of payments.
- Order Tracking Module – This module helps customers track their order status in real time, such as order placed, preparing, and delivered, improving transparency and user experience.

TABLE DESIGN

USER TABLE

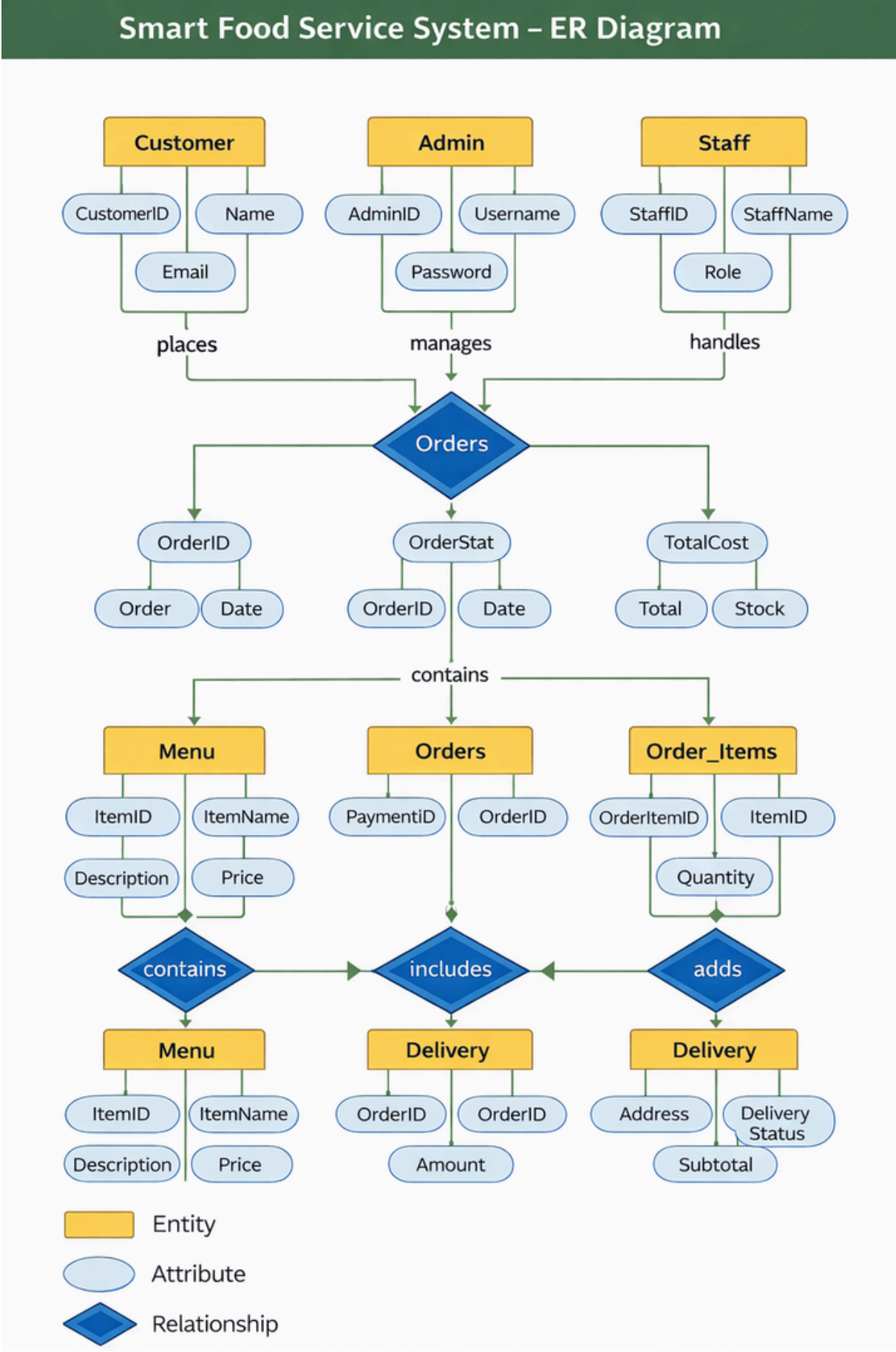
S. No	Field name	Datatype	Constraints
1	id	int	Primary key
2	name	varchar(50)	Not null
3	email	varchar(50)	Not null
4	password	varchar(50)	Not null
5	Role	varchar(50)	Not null
6	Phone	int	Not null
7	Address	varchar(50)	Not null

TABLE DESIGN

MENU TABLE

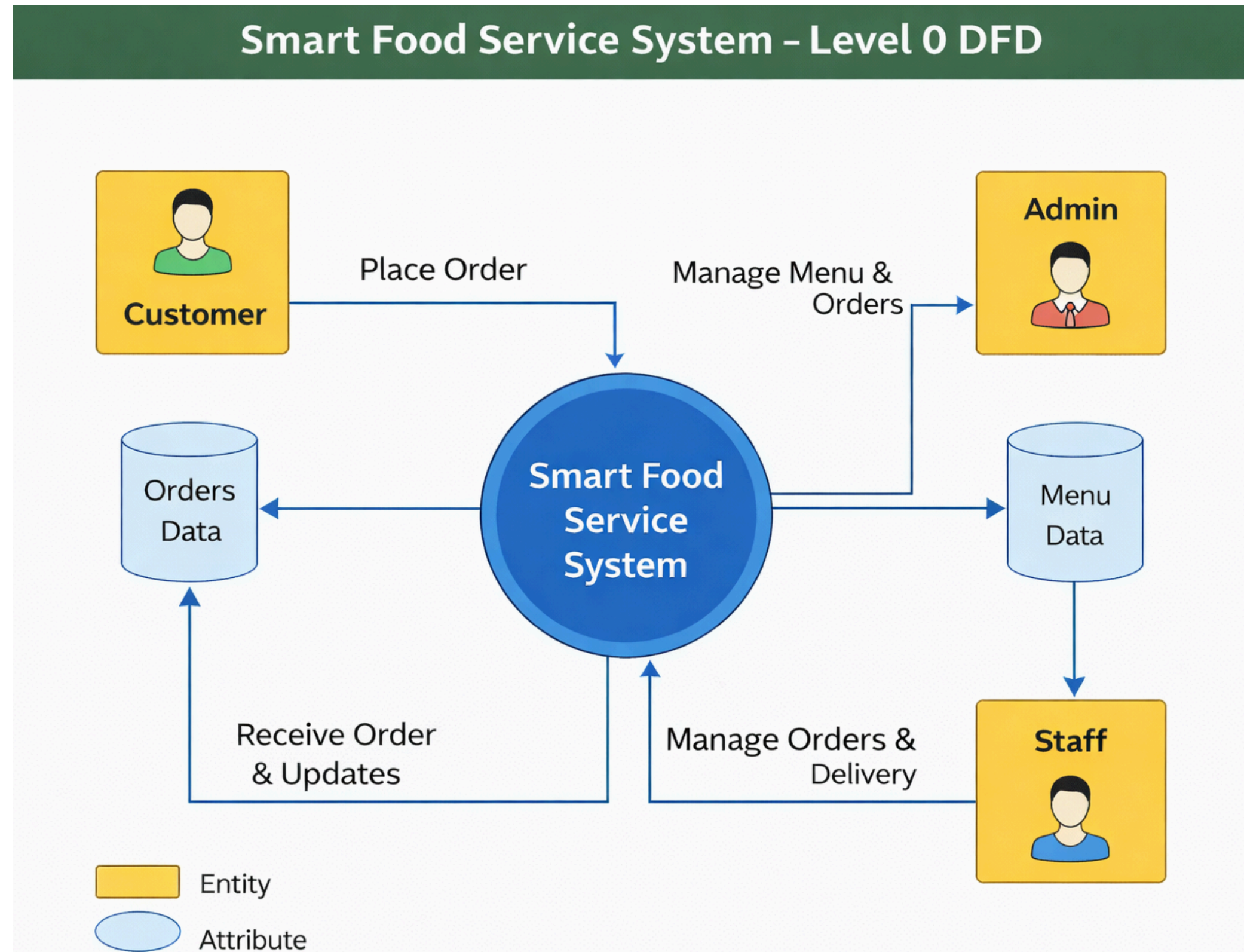
S. No	Field name	Datatype	Constraints
1	Menu_id	int	Primary key
2	Category_Name	Varchar	Foreign Key
3	Description	varchar	Not null
4	Price	int	Not null
5	Image_Url	Varchar	Not null

ER DIAGRAM



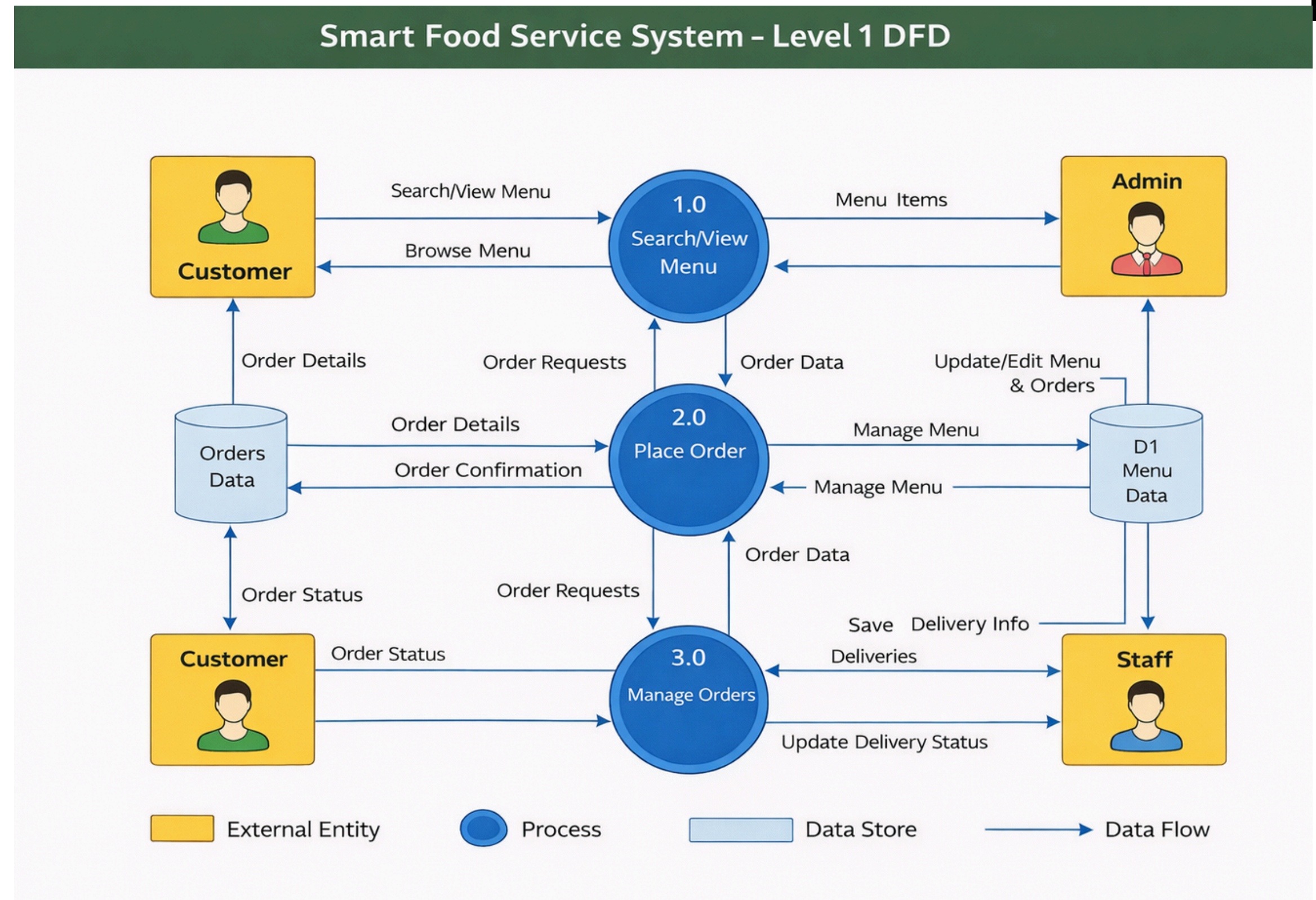
DATA FLOW DIAGRAM

LEVEL 0

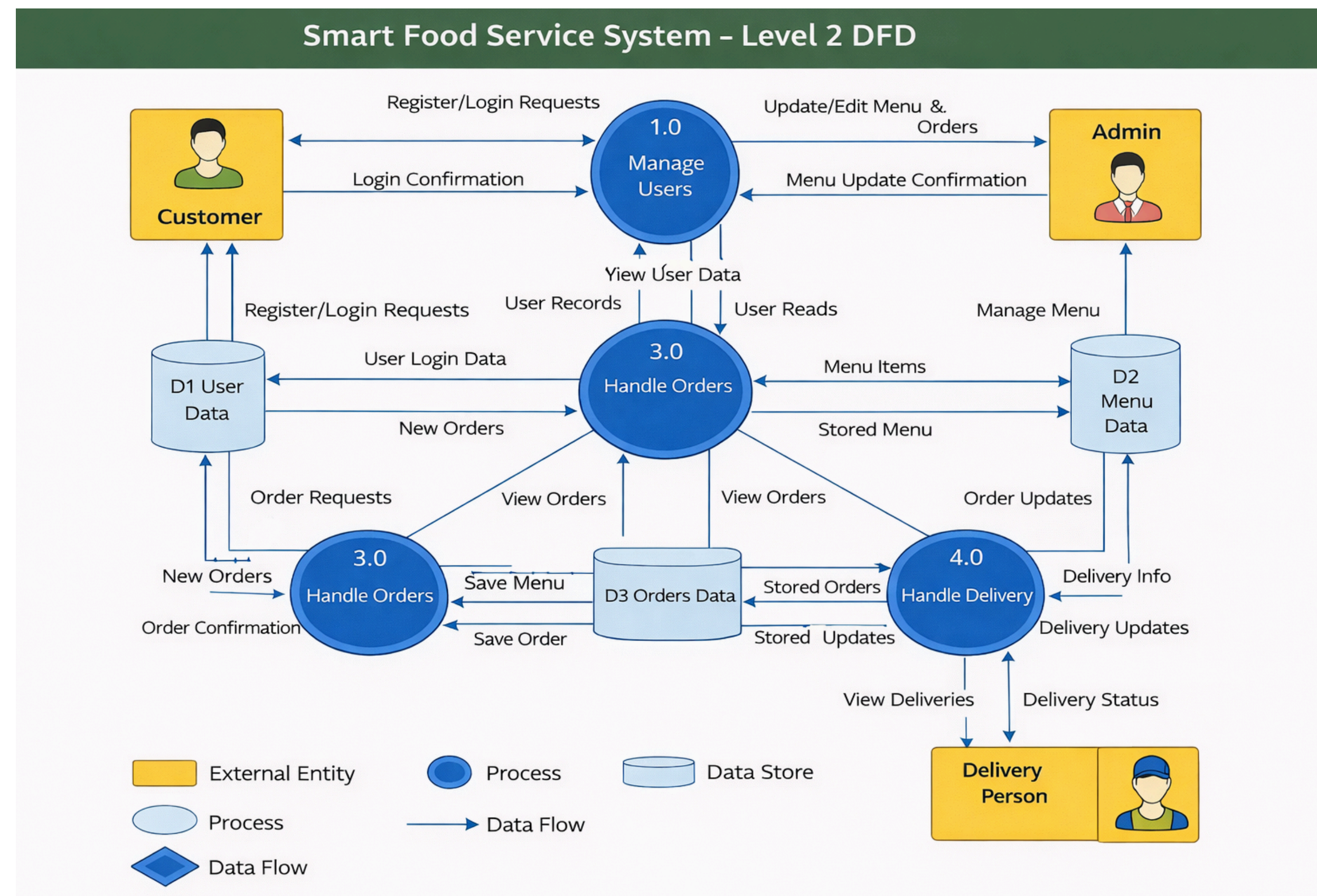


DATA FLOW DIAGRAM

LEVEL 1



DATA FLOW DIAGRAM LEVEL 2

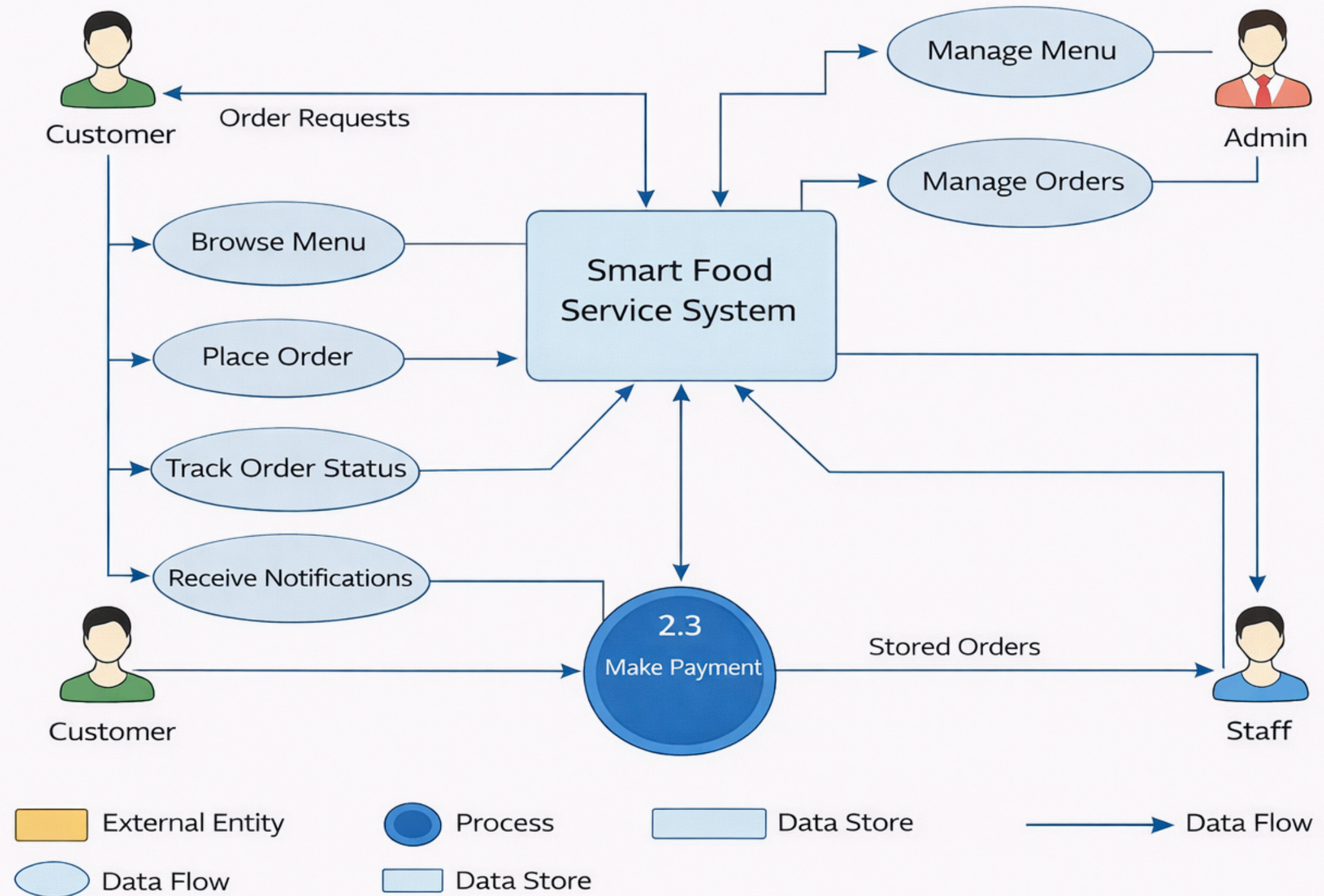


DATA DICTIONARY

Field name	Datatype	Constraints
id	int	Store id
name	varchar(50)	Store name
email	varchar(50)	Store email
password	varchar(50)	Store password
Role	varchar(50)	Defines whether the user is Admin or Customer or Staff
Phone	int	Store the Customer phone for OTP verification
Address	varchar(50)	Store address for delivery order
Menu_id	int	Store menu id
Category_Name	Varchar	Store category name
Description	varchar	Store enrolled description
Price	int	Store enrollment price
Image_Url	Varchar	Store enrollment image url

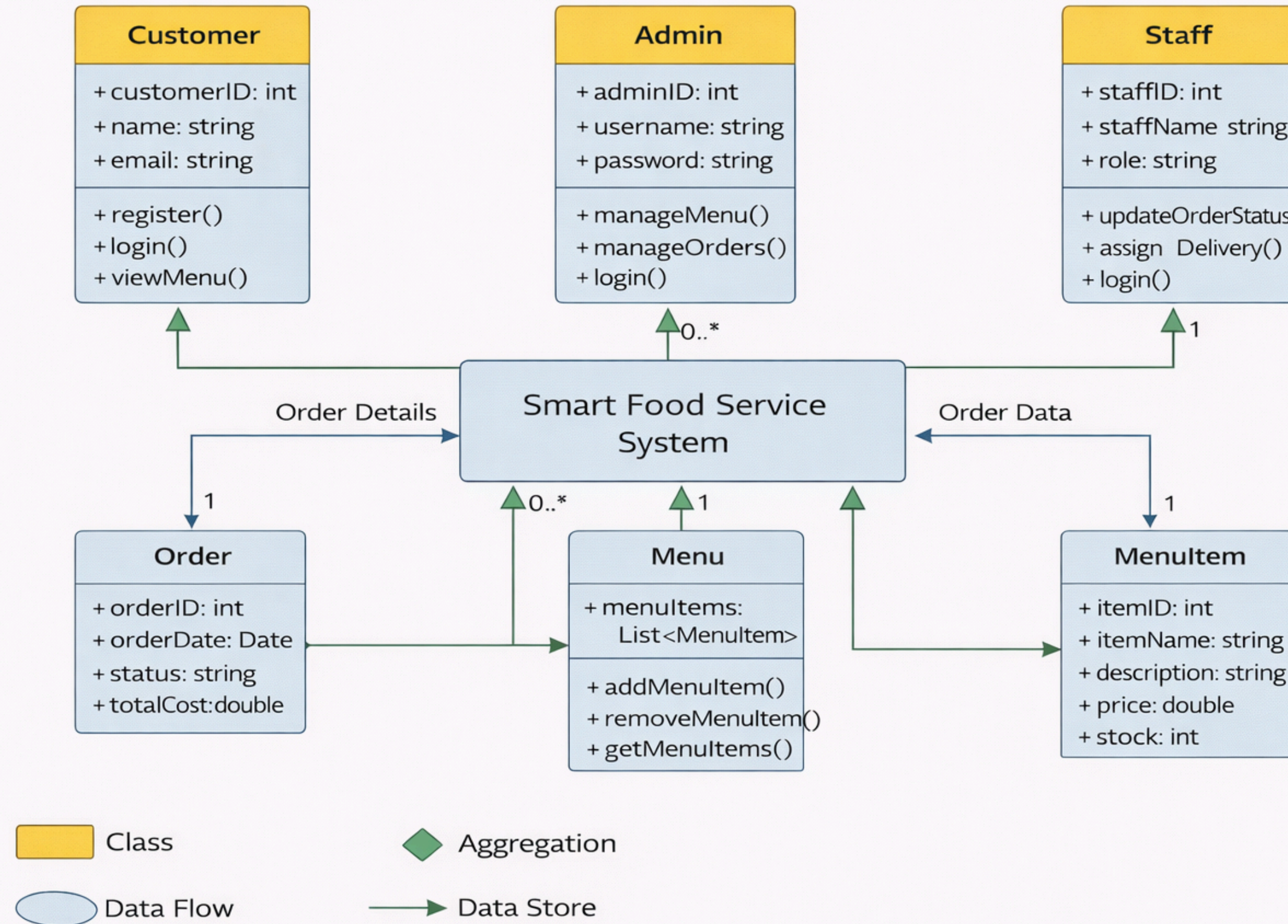
USE CASE DIAGRAM

Smart Food Service System - Use Case Diagram



CLASS DIAGRAM

Smart Food Service System - Class Diagram



SAMPLE CODING



HOME PAGE

```
import React, { useState, useEffect } from 'react';
import { Link } from 'react-router-dom';
import { useDispatch, useSelector } from 'react-redux';
import { fetchMenu } from '../configure/order/menuSlice.js';
import { fetchCategory } from '../configure/category/categorySlice.js';
import "../style/user/Homepage.css";
const RESTAURANTS = [
  { id: 1, name: 'Pizza Hut', rating: 4.2, time: '25-30 mins', cuisine: 'Pizza, Fast Food', image:
'/images/restaurant_food_grid.png', badge: 'Top Rated' },
  { id: 2, name: 'Burger King', rating: 4.5, time: '20-25 mins', cuisine: 'Burgers, American', image:
'/images/restaurant_food_grid.png', badge: 'Fast Delivery' },
  { id: 3, name: 'Paradise Biryani', rating: 4.8, time: '35-40 mins', cuisine: 'Biryani, Indian', image:
'/images/restaurant_food_grid.png', badge: 'Must Try' },
  { id: 4, name: 'Wow! Momo', rating: 4.1, time: '15-20 mins', cuisine: 'Chinese, Snacks', image:
'/images/restaurant_food_grid.png', badge: '' },
  { id: 5, name: 'Starbucks', rating: 4.6, time: '10-15 mins', cuisine: 'Coffee, Desserts', image:
'/images/restaurant_food_grid.png', badge: '' },
  { id: 6, name: 'Subway', rating: 4.3, time: '20-25 mins', cuisine: 'Healthy, Salads', image:
'/images/restaurant_food_grid.png', badge: 'Healthy' },
]
```

k

```
<div className="footer-col">
  <h4>Experience SFSS App</h4>
  <div className="app-btns">
    <a href="#" className="app-btn">
      <FiPlay size={24} />
      <div><span>GET IT ON</span><strong>Google Play</strong></div>
    </a>
    <a href="#" className="app-btn">
      <FiSmartphone size={24} />
      <div><span>Download on the</span><strong>App Store</strong></div>
    </a>
  </div>
</div>
<div className="footer-bottom">
  <p>© 2026 Smart Food Service System. All rights reserved.</p>
  <div style={{ display: 'flex', gap: '30px' }}>
    <span>Privacy</span>
    <span>Terms</span>
    <span>Cookies</span>
  </div>
</div>
</footer>
</div>
);
}
```



CATEGORY PAGE



```
import React, { useState } from 'react';
import { useDispatch, useSelector } from 'react-redux';
import { FiX, FiPlus, FiEdit2, FiTrash2, FiCheck, FiCornerDownRight } from 'react-icons/fi';
import { addCategory, updateCategory, deleteCategory } from '../../configure/category/categorySlice.js';

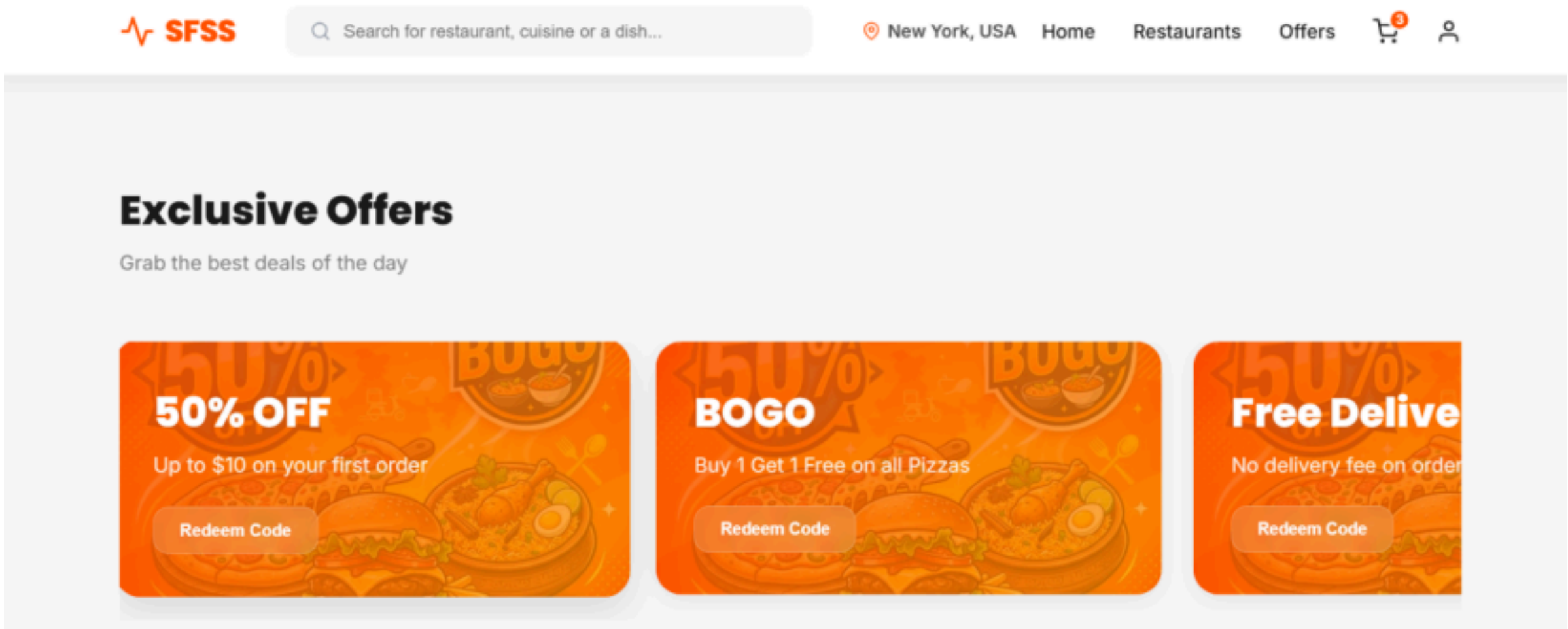
const CategoryManager = ({ onClose }) => {
  const dispatch = useDispatch();
  const { items: categories, loading } = useSelector(state => state.category);

  const [newCategoryName, setNewCategoryName] = useState('');
  const [editingId, setEditingId] = useState(null);
};

const handleUpdate = async (id) => {
  if (!editName.trim()) return;
  try {
    await dispatch(updateCategory({ id, name: editName.trim() })).unwrap();
    setEditingId(null);
    setEditName('');
    setError('');
  } catch (err) {
    setError(err || 'Failed to update category');
  }
};
```

SAMPLE SCREEN

HOME PAGE



MENU PAGE

Menu Management

+ Add Menu Item

4
Total Items

2
Categories

\$
\$410.70
Avg. Price

Search menu items...

All Categories

Showing 4 of 4 items

SOUTH INDIAN



Dindigul Mutton Gravy

Dindigul Mutton Gravy

\$455.56

Edit

Delete

NORTH INDIAN



Gopi Manjuriyan

Gopi Manjuriyan Salad

\$256.77

Edit

Delete

SOUTH INDIAN



Chicken 65

Fired and Crispy foodie

\$500.34

Edit

Delete

Activate Windows
Go to Settings to activate Windows.

SYSTEM TESTING

UNIT TESTING

Unit testing involves testing individual components or modules of the system independently to ensure that each part works correctly. In the Smart Food Service System, unit testing is performed on modules such as user login, menu management, order processing, and payment handling.

INTEGRATION TESTING

Integration testing is carried out to verify the interaction between different modules of the system. After unit testing, individual components are combined and tested as a group. In this system, integration testing ensures that modules such as the frontend interface, backend API, and database work together seamlessly.

VALIDATION TESTING

In this system, validation testing is performed by providing different inputs and verifying whether the outputs are correct and meaningful.

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HARDWARE AND SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS

- Processor: Intel i7 or above
- Laptop or Desktop
- RAM: 16 GB
- Hard Disk: 256 GB free space
- Internet connection

SOFTWARE REQUIREMENTS

- Frontend: React.js
- Backend: C#
- Database: MYSQL
- Code Editor: VS Code
- Browser: Chrome / Edge



CONCLUSION

The Smart Food Service System successfully provides a digital solution to simplify and enhance the food ordering process in restaurants. The system allows customers to conveniently browse menus, place orders, and make payments online, reducing waiting time and improving overall user experience. From the administrative perspective, the system offers efficient management of menu items, categories, users, and orders through a centralized platform.

FUTURE ENHANCEMENTS

- Develop an Android/iOS mobile app to provide better accessibility and user experience.
- Implement intelligent suggestions based on user preferences and order history.
- Enhance tracking by integrating live location features for delivery services.

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PUBLICATON DETAILS

JOURNALPAPER

AI - Powered Mental Health Risk Assessment For Schizophrenia

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This is to certify that ARUN GLADSON S from francis xavier engineering college has successfully submitted and presented the research paper titled AI - Powered Mental Health Risk Assessment For Schizophrenia virtually at the 2nd International Conference on Recent Innovation in Life Sciences, Healthcare for Sustainable Development (ICRILSH-2026) organized by Department of MCA, Francis Xavier Engineering College, India and IFERP Life Sciences-Formerly Bioleagues, held on 26th & 27th February 2026 in Kuala Lumpur, Malaysia as a Hybrid Conference. Accredited by CPD, the conference offers 16 credit hours of professional development.

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THANK YOU